

Lab 17: RNA-Seq Quantification

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RNA-Seq Quantification and PCA Analysis

First, we load the required libraries and import the abundance data using `tximport`:

```
setwd("/Users/omar")
library(tximport)

folders <- dir(pattern="SRR21568*")
files <- file.path(folders, "abundance.h5")
names(files) <- sub("_quant", "", folders)

txi <- tximport(files, type="kallisto", txOut=TRUE)
```

Filtering

Remove transcripts with no reads and no variation:

```
to.keep <- rowSums(txi$counts) > 0
kset.nonzero <- txi$counts[to.keep,]

keep2 <- apply(kset.nonzero, 1, sd) > 0
x <- kset.nonzero[keep2,]
```

PCA

```
pca <- prcomp(t(x), scale=TRUE)
summary(pca)
```

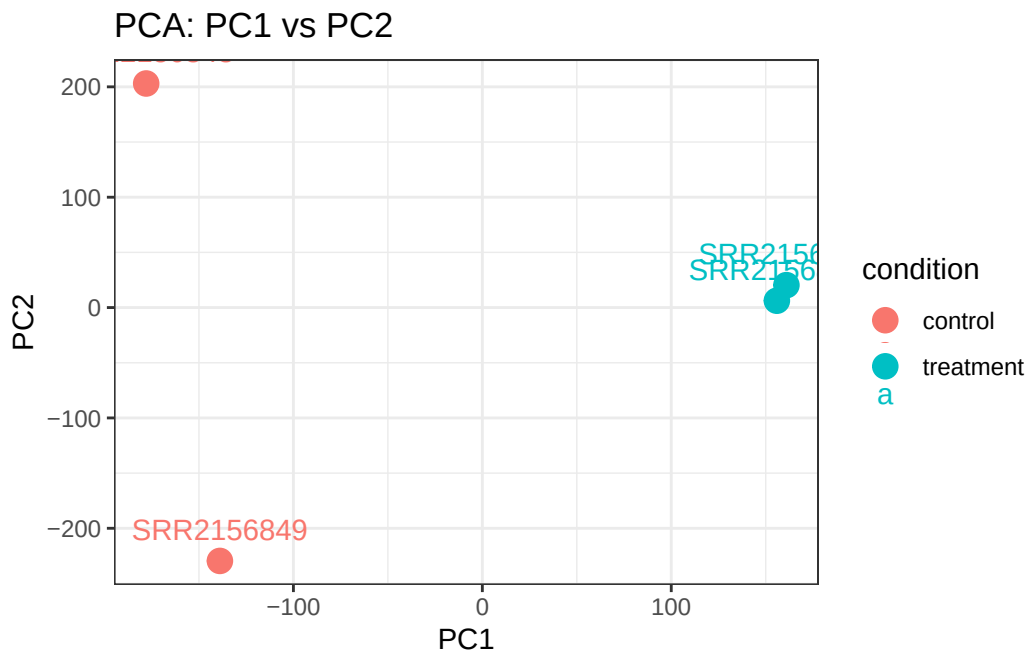
Importance of components:

	PC1	PC2	PC3	PC4
Standard deviation	183.6379	177.3605	171.3020	1e+00
Proportion of Variance	0.3568	0.3328	0.3104	1e-05
Cumulative Proportion	0.3568	0.6895	1.0000	1e+00

```
library(ggplot2)

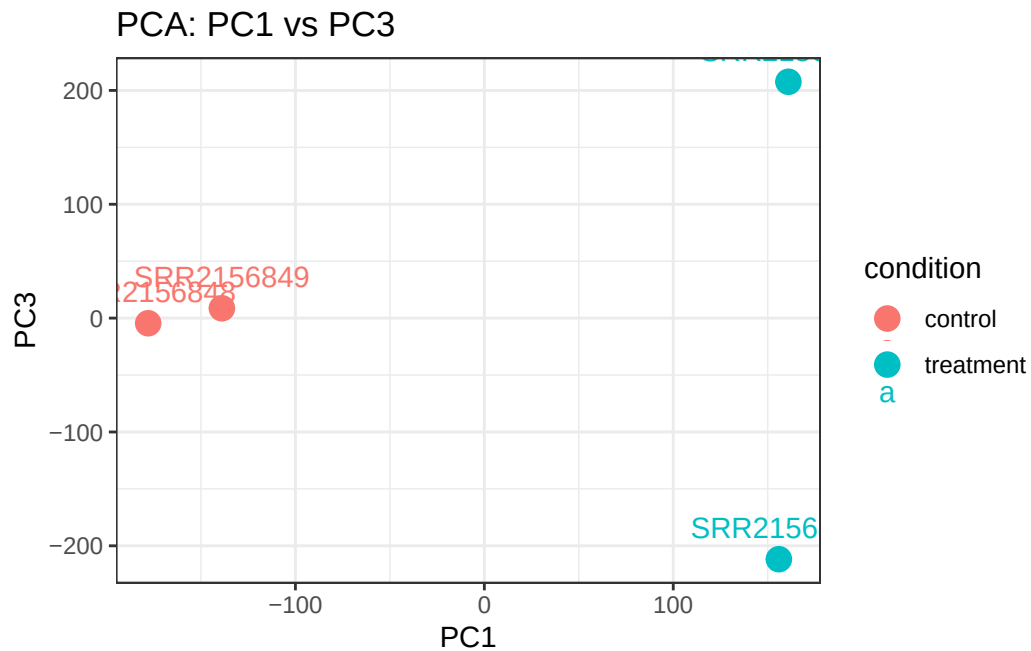
pca_df <- as.data.frame(pca$x)
pca_df$sample <- rownames(pca_df)
pca_df$condition <- c("control", "control", "treatment", "treatment")

# PC1 vs PC2
ggplot(pca_df, aes(x=PC1, y=PC2, color=condition, label=sample)) +
  geom_point(size=4) +
  geom_text(vjust=-1) +
  labs(title="PCA: PC1 vs PC2") +
  theme_bw()
```

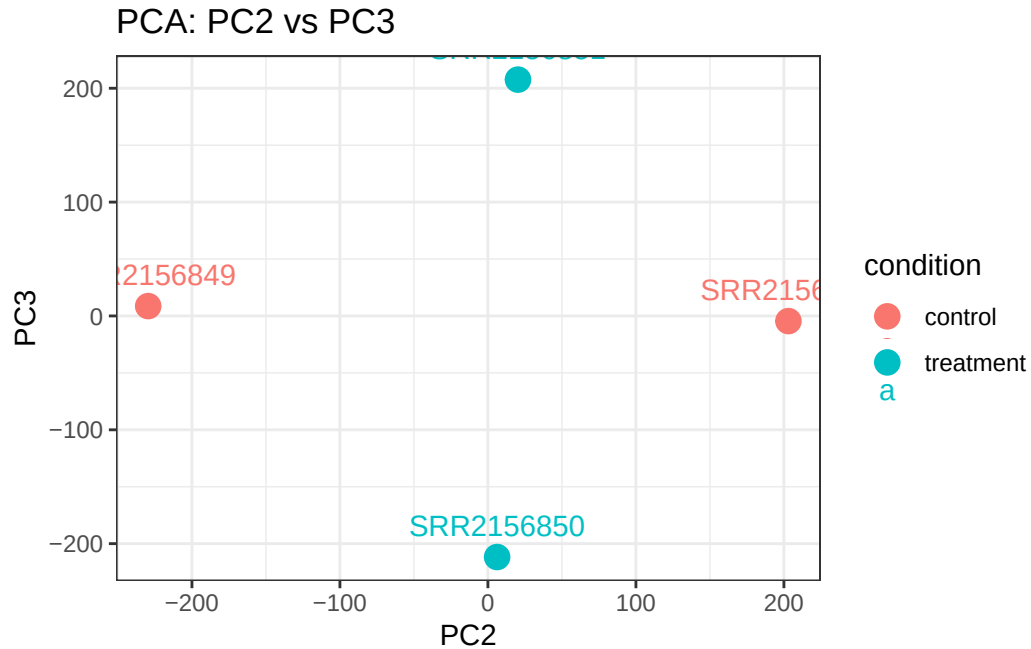


```
# PC1 vs PC3
ggplot(pca_df, aes(x=PC1, y=PC3, color=condition, label=sample)) +
  geom_point(size=4) +
```

```
geom_text(vjust=-1) +
labs(title="PCA: PC1 vs PC3") +
theme_bw()
```



```
# PC2 vs PC3
ggplot(pca_df, aes(x=PC2, y=PC3, color=condition, label=sample)) +
  geom_point(size=4) +
  geom_text(vjust=-1) +
  labs(title="PCA: PC2 vs PC3") +
  theme_bw()
```



Lab Questions

Q: What do the abundance.tsv files contain?

Each file contains columns for transcript ID, length, effective length, estimated counts, and TPM. Most transcripts have zero counts — only a subset are expressed in each sample.

Q: PCA Interpretation

PC1 separates the two control samples (blue) from the two CRISPR treatment samples (red), indicating a strong difference in transcript abundance between conditions. PC2 separates the two control replicates from each other, and PC3 separates the two treatment replicates.